



Figure 4: Log in screen of XigmaNAS

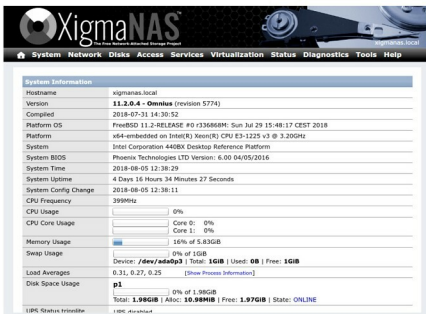


Figure 5: Home page of XigmaNAS

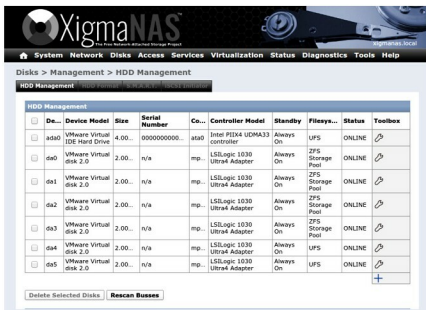


Figure 6: Disk management from a single screen - XigmaNAS

Differences between NAS and SAN

The key difference between NAS and SAN storage is that NAS provides file-level access and SAN is a block storage solution. In NAS, data transmission occurs via Ethernet or TCP/IP protocols, whereas SAN uses fibre channel or IP protocols. NAS storage

uses the existing LAN for network infrastructure and SAN uses a separate dedicated network infrastructure. SAN storages are usually used for servers, unlike NAS. SAN looks like a local disk, and diskless servers are possible while using it. While SAN doesn't offer metadata support, NAS does. SAN is costlier than NAS and is more complex. The various NAS protocols are common Internet file system (CIFS), server message block (SMB) and network file system (NFS). SAN protocols are fibre channel (FC), Internet small computer systems interface (iSCSI), fibre channel over Ethernet (FCoE), fibre channel over Ethernet (FCoE) and non-volatile memory express over fabric (NVMeoF). Modern storage systems support both NAS and SAN.

Popular open source NAS and SAN solutions

While selecting a storage type, several factors are considered, such as budget, type of data, scaling considerations and usage patterns. To design a cost-efficient storage system using open source, a solution that suits one's unique use-case must be chosen.

Many options are available in the open source world for implementing NAS and SAN. A few of these are – FreeNAS (on Windows, Linux, Mac), XigmaNAS (on Windows, Linux, Mac), OpenMediaVault (on Debian Linux), Rockstor (on CentOS Linux), Openfiler Community Edition (on Linux), Amahi (on Fedora and Ubuntu), PetaSAN and CryptoNAS.

XigmaNAS: An embedded open source solution for designing NAS

XigmaNAS can be installed on a wide range of hardware, across multiple operating systems (Windows, Apple, UNIX-like OS), over a number of standard network protocols. It can be installed on a CD/USB/SSD key, hard disk or booted from a LiveCD/ LiveUSB with a small USB key for config storage.